8. Construct a C Program to Simulate Round Robin Scheduling Algorithm with C

Code - Part 1 (Lines 1–35)

```c
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, time = 0, remain, flag = 0, quantum;
6      int at[10], bt[10], rt[10];
7      int wt = 0, tat = 0;
8
9      printf("Enter number of processes: ");
10     scanf("%d", &n);
11
12     remain = n;
13
14     for(i = 0; i < n; i++)
15     {
16         printf("Enter Arrival Time and Burst Time for P[%d]: ", i+1);
17         scanf("%d %d", &at[i], &bt[i]);
18         rt[i] = bt[i]; // remaining time
19     }
20
21     printf("Enter Time Quantum: ");
22     scanf("%d", &quantum);
23
24     printf("\nProcess\tTAT\tWaiting Time\n");
25
26     for(time = 0, i = 0; remain != 0;)
27     {
28         if(rt[i] <= quantum && rt[i] > 0)
29         {
30             time += rt[i];
31             rt[i] = 0;
32             flag = 1;
33         }
34         else if(rt[i] > 0)
35         {
```

Code - Part 2 (Lines 36–68)

```
36              rt[i] -= quantum;
37              time += quantum;
38          }
39
40          if(rt[i] == 0 && flag == 1)
41          {
42              remain--;
43
44              printf("P[%d]\t%d\t%d\n",
45                     i+1,
46                     time - at[i],
47                     time - at[i] - bt[i]);
48
49              wt += time - at[i] - bt[i];
50              tat += time - at[i];
51
52              flag = 0;
53          }
54
55          // move to next process safely
56          if(i == n - 1)
57              i = 0;
58          else if(at[i+1] <= time)
59              i++;
60          else
61              i = 0;
62      }
63
64      printf("\nAverage Turnaround Time: %.2f", (float)tat/n);
65      printf("\nAverage Waiting Time: %.2f\n", (float)wt/n);
66
67      return 0;
68  }
```

Output

```
Enter number of processes: 3
Enter Arrival Time and Burst Time for P[1]: 0
5
Enter Arrival Time and Burst Time for P[2]: 1
3
Enter Arrival Time and Burst Time for P[3]: 2
1
Enter Time Quantum: 2

Process TAT     Waiting Time
P[3]    3       2
P[2]    7       4
P[1]    9       4

Average Turnaround Time: 6.33
Average Waiting Time: 3.33


...Program finished with exit code 0
Press ENTER to exit console.
```